

MAYAC10761

## Butadiene sulfone

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 3-环丁烯砜  
**Butadiene sulfone**

**Cat No. :** AC10761R3  
**Synonyms** 3-Sulfolene; 2,5-Dihydrothiophene-1,1-dioxide  
**CAS No** 77-79-2  
**Molecular Formula** C4 H6 O2 S

**Supplier** **UK entity/business name**  
Thermo Fisher Scientific (Heysham),  
Shore Road,  
Port of Heysham Industrial Park,  
Heysham, Lancashire, LA3 2XY  
United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
Off-white

**Odor**  
Stench

#### Emergency Overview

May be harmful if swallowed. Causes serious eye damage. Stench.

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 5 |
| Serious Eye Damage/Eye Irritation | Category 1 |

#### Label Elements



**Signal Word****Danger****Hazard Statements**

H303 - May be harmful if swallowed

H318 - Causes serious eye damage

**Precautionary Statements****Prevention**

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May be harmful if swallowed. Corrosive. Causes eye burns.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Stench. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component   | CAS No  | Weight % |
|-------------|---------|----------|
| 3-Sulfolene | 77-79-2 | >95      |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable. Causes eye burns.

**Self-Protection of the First Aider**

Use personal protective equipment as required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls**

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**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                 |                               |                                          |
|---------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>               | Off-white                     |                                          |
| <b>Physical State</b>           | Solid                         |                                          |
| <b>Odor</b>                     | Stench                        |                                          |
| <b>Odor Threshold</b>           | No data available             |                                          |
| <b>pH</b>                       | Not applicable                |                                          |
| <b>Melting Point/Range</b>      | 63 - 66 °C / 145.4 - 150.8 °F |                                          |
| <b>Softening Point</b>          | No data available             |                                          |
| <b>Boiling Point/Range</b>      | No information available      |                                          |
| <b>Flash Point</b>              | 112 °C / 233.6 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available      |                                          |
| <b>Explosion Limits</b>         | No data available             |                                          |

**SAFETY DATA SHEET****Butadiene sulfone**

|                                                |                          |       |
|------------------------------------------------|--------------------------|-------|
| <b>Vapor Pressure</b>                          | No data available        |       |
| <b>Vapor Density</b>                           | Not applicable           | Solid |
| <b>Specific Gravity / Density</b>              | No data available        |       |
| <b>Bulk Density</b>                            | No data available        |       |
| <b>Water Solubility</b>                        | Soluble                  |       |
| <b>Solubility in other solvents</b>            | No information available |       |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |       |
| <b>Component</b>                               | <b>log Pow</b>           |       |
| 3-Sulfolene                                    | -0.45                    |       |
| <b>Autoignition Temperature</b>                | No data available        |       |
| <b>Decomposition Temperature</b>               | No data available        |       |
| <b>Viscosity</b>                               | Not applicable           | Solid |
| <b>Explosive Properties</b>                    | No information available |       |
| <b>Oxidizing Properties</b>                    | No information available |       |
| <b>Molecular Formula</b>                       | C4 H6 O2 S               |       |
| <b>Molecular Weight</b>                        | 118.16                   |       |

**SECTION 10. STABILITY AND REACTIVITY**

|                                 |                                                           |
|---------------------------------|-----------------------------------------------------------|
| <b>Stability</b>                | No information available.                                 |
| <b>Hazardous Reactions</b>      | None under normal processing.                             |
| <b>Hazardous Polymerization</b> | No information available.                                 |
| <b>Conditions to Avoid</b>      | Incompatible products. Excess heat. Avoid dust formation. |
| <b>Materials to avoid</b>       | Strong oxidizing agents.                                  |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component   | LD50 Oral                 | LD50 Dermal        | LC50 Inhalation |
|-------------|---------------------------|--------------------|-----------------|
| 3-Sulfolene | LD50 = 2547 mg/kg ( Rat ) | >5 g/kg ( Rabbit ) |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

|                                            |                           |
|--------------------------------------------|---------------------------|
| (h) STOT-single exposure;                  | No data available         |
| (i) STOT-repeated exposure;                | No data available         |
| Target Organs                              | No information available. |
| (j) aspiration hazard;                     | Not applicable<br>Solid   |
| Symptoms / effects, both acute and delayed | No information available  |

## SECTION 12. ECOLOGICAL INFORMATION

|                     |                                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|

| Component   | Freshwater Fish     | Water Flea          | Freshwater Algae | Microtox |
|-------------|---------------------|---------------------|------------------|----------|
| 3-Sulfolene | LC50 = 940 mg/L 96h | EC50 = 800 mg/L 48h |                  |          |

|                               |                                                                            |
|-------------------------------|----------------------------------------------------------------------------|
| Persistence and Degradability | Not readily biodegradable                                                  |
| Persistence                   | Soluble in water, Persistence is unlikely, based on information available. |

|                           |                             |
|---------------------------|-----------------------------|
| Bioaccumulative Potential | Bioaccumulation is unlikely |
|---------------------------|-----------------------------|

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| 3-Sulfolene | -0.45   | No data available             |

|                  |                                                                                                                                                           |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobility in soil | The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                     |                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point.                                                                                              |
| Other Information                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.                   |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                         |             |
|-------------------------|-------------|
| Technical Shipping Name | 3-Sulfolene |
|-------------------------|-------------|

|          |               |
|----------|---------------|
| IMDG/IMO | Not regulated |
|----------|---------------|

**SAFETY DATA SHEET****Butadiene sulfone****Technical Shipping Name** 3-Sulfolene**IATA**

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.  
**Technical Shipping Name** 3-Sulfolene  
**Hazard Class** 9  
**Packing Group** III

**Special Precautions for User** No special precautions required**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component   | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 3-Sulfolene | -                                                   | -                                       | X    | -     | 201-059-7 | X    | X   | X     | X    | X    | -    | KE-10754 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Creation Date** 15-Dec-2011  
**Revision Date** 21-Aug-2023  
**Revision Summary** SDS sections updated, 1, 2, 9, 11, 12, 15, 16.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend****CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%

## Butadiene sulfone

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**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**OECD** - Organisation for Economic Co-operation and Development**BCF** - Bioconcentration factor**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**MARPOL** - International Convention for the Prevention of Pollution from Ships**ATE** - Acute Toxicity Estimate**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**